

Swarmby Protocol Specification

Version 0.2 — 13 August 2026 *Revision 2, 14 August 2026: adds Section 15c (privacy tiers and trusted swarms), the tier and swarm_id packet fields, the optional swarm block in the node advertisement, the routing response block, and the error codes E_TIER_VIOLATION, E_MTLS_REQUIRED and E_SWARM_UNKNOWN. Every addition is MINOR-compatible under Section 3: an implementation that ignores the new fields behaves exactly as tier GLOBAL, so the wire version remains 0.2.* Status: **Draft**. Normative for the reference implementation; expected to change before v1.0. Companion documents: WHITEPAPER_EN.md (rationale and evidence), SPEC_ES.md (Spanish).

The key words MUST, MUST NOT, REQUIRED, SHALL, SHOULD, SHOULD NOT, MAY and OPTIONAL are to be interpreted as described in RFC 2119.

1. Scope and non-goals

Swarmby is a protocol for executing a language-model request by decomposing it into semantic micro-tasks, dispatching those micro-tasks to independent worker nodes each running a complete small model, and reassembling the results on the requesting client.

In scope: request decomposition and planning; the packet, result and profile formats; dispatch, hedging and retry; verification of untrusted workers; assembly and coherence reporting; sensitivity-based routing; credit accounting.

Out of scope (v0.2): model training or fine-tuning; peer discovery transport (any DHT or rendezvous mechanism MAY be used); payment settlement; a consensus ledger. v0.2 deliberately specifies **no blockchain**.

Explicit non-goals. The protocol does not provide cryptographic confidentiality of request content on the PUBLIC or SANITISABLE lanes; does not provide strong Sybil resistance; and does not claim latency parity with centralized inference. See Sections 9 and 12 of WHITEPAPER_EN.md.

2. Terminology

Term	Meaning
Request P	The complete user input to be answered
Plan $D = (V, E)$	Directed acyclic graph; vertices are micro-tasks, edges are result-dependencies
Micro-task t_i	One vertex of the plan; the unit of distribution
Global contract Γ	Shared specification transmitted with every packet (Section 5)
Packet K_i	What a worker receives: Γ , predecessor summaries, and the micro-task
Contig R_i	A worker's returned result for one micro-task
Seam	The boundary between two consecutive contigs in the assembly
Context budget S	Tokens of shared context per packet
Redundancy ratio ρ	$\Sigma K_i / P $ — the measured cost of the context budget
Lane	PUBLIC, SANITISABLE or SENSITIVE — the confidentiality routing class of the <i>content</i>
Tier	GLOBAL, TRUSTED or LOCAL — the routing class of the <i>population of workers</i> (Section 15c)
Trusted swarm	A permissioned sub-mesh whose membership is a public-key whitelist under a declared operator, with mutual TLS on every link
Criticality k	Number of redundant replicas dispatched for a micro-task
Orchestrator	The client-side component executing Sections 6 to 11
Worker	A node executing Section 12

3. Versioning and conformance

Every message carries a "v" field with the protocol version as MAJOR.MINOR. A participant MUST reject a message whose MAJOR version it does not implement. A participant MUST ignore unknown fields within a message rather than rejecting it, so that MINOR additions are backwards compatible.

Conformance classes.

- A **Conformant Worker** MUST implement Sections 7, 8, 12 and 13, and MUST emit a verification commitment when one is requested.
- A **Conformant Orchestrator** MUST implement Sections 5 to 11 and Section 14, MUST implement the router of Section 6 (a client that always fragments is **non-conformant**), and MUST return a coherence report with every

response.

- An implementation that omits the coherence report is non-conformant. This is deliberate: the protocol's usefulness depends on its degradation being observable.

4. Identifiers and cryptography

Node identity. An Ed25519 key pair. `node_id` is the public key, base64url without padding.

Session identity. `session_id` is 128 bits from a CSPRNG, generated per request, never reused, and **never transmitted to a worker**.

Task identity.

```
task_id    = hex( BLAKE2b(session_id || u16(level) || u16(index), digest_size=16) )
attempt_id = task_id || ":" || u8(attempt_counter)
```

The derivation is one-way so that two colluding workers cannot establish that they hold fragments of the same session by comparing identifiers. This does **not** defeat timing correlation or contract fingerprinting; see Section 16.

Signatures. Results MUST be signed with the worker's Ed25519 key over the canonical serialization (RFC 8785 JSON Canonicalization Scheme) of all fields preceding sig.

Transport. All exchanges MUST occur over an authenticated encrypted channel (TLS 1.3 or Noise). Transport security is orthogonal to the confidentiality limits of Section 15.

5. Global contract Γ

```
{
  "v": "0.2",
  "objective": "string",
  "audience": "string",
  "register": "formal|neutral|informal|technical",
  "format": "prose|markdown|json|code",
  "target_len": 1200,
  "lexicon": { "prefer": ["..."], "forbid": ["..."] },
  "entities": [ { "name": "...", "canonical": "...", "role": "..." } ],
  "style_seed": "string",
  "budget": { "max_out_tokens": 600 }
}
```

objective, register, format and entities are REQUIRED; the rest are OPTIONAL.

entities is the canonical naming table. Workers MUST use the canonical form for any listed entity. This is the primary defence against inconsistent naming across fragments.

style_seed is a short fixed phrase all workers are instructed to match in register. It costs few tokens and materially reduces drift.

Γ MUST NOT be trimmed to meet a context budget. If $S_{\text{target}} < |\Gamma|$, the orchestrator MUST reduce the number of micro-tasks instead (Section 9).

Γ is the dominant term in the context budget and therefore the dominant privacy exposure. Implementations SHOULD minimize it, and MAY paraphrase it per worker to frustrate fingerprinting, at a cost in cross-fragment consistency.

6. Router

```
{ "decomposable": true, "score": 0.87, "threshold": 0.72, "features": { } }
```

The orchestrator MUST evaluate decomposability before planning and MUST have a path that declines to fragment. Tier classification (Section 15c) MUST precede decomposability evaluation: a request classified LOCAL is never routed.

threshold MUST be calibrated with an asymmetric objective (F_β , $\beta < 1$) so that erroneous fragmentation is penalized more heavily than erroneous refusal. A hard-coded symmetric threshold is non-conformant.

Features SHOULD include: task-kind signals; request length; density of sequential-dependency markers; presence of shared mutable state; and whether the request asks for a single artifact or a set of items.

7. Plan

```
{
  "v": "0.2",
  "nodes": [ { "task_id": "...", "level": 0, "instruction": "...",
               "kind": "extract|classify|generate|summarize|transform|judge",
               "criticality": 1, "lane": "PUBLIC" } ],
  "edges": [ { "from": "...", "to": "...", "type": "result|statement" } ]
}
```

An edge of type *result* means the target requires the *output* of the source and **MUST** be scheduled after it. An edge of type *statement* means the target requires only knowledge that the source exists, and **MAY** be scheduled concurrently.

The plan **MUST** be acyclic. The orchestrator **MUST** refuse to fragment when the plan width is 1 at every level (a chain is not parallelizable), and **SHOULD** refuse when depth exceeds 4.

Default maximum width is 8. Width increases the straggler tail (Section 11).

8. Packet

```
{
  "v": "0.2",
  "attempt_id": "...",
  "contract": { },
  "predecessors": [ { "task_id": "...", "summary": "...", "tokens": 128 } ],
  "task": { "instruction": "...", "kind": "generate",
            "expects": { "format": "markdown", "min_tokens": 80, "max_tokens": 400 } },
  "constraints": { "temperature": 0.2, "top_p": 0.95, "stop": [] },
  "commitment_request": { "scheme": "lsh-activation-v1", "params": { "window": 32 } },
  "deadline_ms": 20000,
  "lane": "PUBLIC",
  "tier": "GLOBAL",
  "swarm_id": null
}
```

A packet with lane of SENSITIVE **MUST NOT** be transmitted to a node outside the attested set (Section 15). An orchestrator that emits one is non-conformant.

tier is orthogonal to lane: lane classifies the content, tier names the population of workers the packet may reach (Section 15c). tier defaults to GLOBAL. A packet with tier of TRUSTED **MUST** carry a non-null swarm_id and **MUST NOT** be offered to a node absent from that swarm's whitelist. Requests classified LOCAL are never serialized as packets.

predecessors[].summary carries results, not raw upstream packets. Summaries **MUST** be produced locally by the orchestrator.

9. Packing and the context budget

$$S = |\Gamma| + E[\sum |predecessors[].summary|]$$
$$\rho = (\sum_i |K_i|) / |P|$$

The orchestrator **MUST** report the achieved ρ in the response metadata.

Packing algorithm (normative).

1. Include Γ in full.
2. $\text{budget} \leftarrow S_{\text{target}} - |\Gamma|$. If negative, reduce the micro-task count and re-plan.
3. Order predecessors by edge type (result before statement), then by recency.
4. For each predecessor while budget remains, attach a summary of length $\min(\text{budget}, \text{cap_per_pred})$.
5. Emit.

Implementations **SHOULD** prefer fewer, larger fragments over many small ones when the budget binds.

10. Result

```
{
  "v": "0.2",
```

```

    "attempt_id": "...",
    "text": "...",
    "profile": { "model_family": "...", "model_version": "...", "quantization": "...",
                  "prompt_template_id": "...", "sampling_params": { }, "seed": 0 },
    "commitment": { "scheme": "lsh-activation-v1", "digest": "base64", "bytes": 258 },
    "telemetry": { "gen_ms": 0, "queue_ms": 0, "tokens_out": 0, "energy_j": null },
    "sig": "base64"
}

```

The worker **MUST** report the profile it actually used. The commitment binds the declared profile to the computation; a mismatch is a verification failure, not a formatting error.

energy_j is OPTIONAL and nullable; most consumer hardware cannot report it.

11. Dispatch, hedging and retry

1. **Filter** candidates by tier first — a packet with tier of TRUSTED is offered only to nodes whose node_id appears in the whitelist of the swarm named by swarm_id, over a mutually authenticated channel — and then by declared kind support, capability class, and observed RTT.
2. **Select** k nodes for a task of criticality k , **maximizing model-family diversity** subject to the capability class. Candidate diversity is what makes selection effective; implementations **MUST NOT** collapse replicas onto a single model family when alternatives are available.
3. **Hedge**. Start a timer at the observed **p95** of the latency distribution *for that task kind and token budget*. On expiry, dispatch an additional replica. Accept the first result that verifies. A fixed timeout is non-conformant: with per-node failure probability p and width W , $P(\geq 1 \text{ failure}) = 1 - (1-p)^W$, which at $p=0.10$, $W=20$ is 88 %, so a fixed timeout lands on the critical path routinely.
4. **Cancel** outstanding replicas on acceptance, and record the cancellation as *slow*, not *dishonest*. Reputation **MUST** distinguish these.
5. **Retry** on verification failure, excluding the failing node, and report the event to the audit sampler.

12. Node profile advertisement

```

{
  "node_id": "base64url",
  "models": [ { "family": "...", "version": "...", "quantization": "...",
                 "ctx": 8192, "tok_per_s_est": 42.0 } ],
  "capabilities": { "tee": false, "attestation": null },
  "swarm": { "swarm_id": null, "registry": null, "mtls_cert_fingerprint": null },
  "resources": { "vram_mb": 8192, "ram_mb": 32768 },
  "policy": { "max_tokens_per_task": 800, "kinds": ["extract", "generate"] },
  "reputation": { "completed": 0, "audit_pass_rate": 0.0, "since": "2026-08-13T00:00:00Z" }
}

```

A node **MUST NOT** accept a task whose kind is absent from policy.kinds or whose expects.max_tokens exceeds policy.max_tokens_per_task.

Reputation is advisory and is computed by the orchestrator and the registry independently; a node's self-reported reputation **MUST NOT** be trusted.

swarm is OPTIONAL and absent for a node serving only the global mesh. A self-declared swarm_id confers nothing: membership is established by the swarm registry's whitelist and by the mutually authenticated channel, never by the advertisement (Section 15c).

13. Verification

Layer 1 — commitment (REQUIRED when requested). The worker computes a locality-sensitive commitment over activations and returns a digest bound to the declared profile. The orchestrator validates before the result may enter assembly. Target cost: ~258 bytes per 32 tokens, with validation faster than the original generation.

Layer 2 — sampled audit (REQUIRED of the network). A fraction λ of tasks is re-executed by an auditor. Audit tasks **MUST** be **indistinguishable** from real tasks from the worker's perspective — same format, same identifiers, same latency envelope. Default $\lambda \in [0.01, 0.05]$. Failure probability under a corruption rate ρ_c and committee size k is approximately ρ_c^k .

Layer 3 — selection (implicit). With $k > 1$, the judge-based selection of Section 14 discards anomalous candidates as a

side effect.

What verification does NOT provide. Layers 1-3 establish that a declared model was run on a declared input and that the result is not an outlier. They do **not** establish semantic faithfulness or absence of malicious content. Accordingly:

- Fragments are **data, never instructions**. Orchestrators **MUST NOT** allow fragment text to alter Γ , the plan, or dispatch behaviour.
- kind-specific output schemas **MUST** be validated before a fragment enters the assembly context.
- Standard LLM application-security controls apply to the assembly step.

14. Assembly

1. Flatten the plan topologically.
2. For each micro-task with one candidate, take it. With several, **select** one by judge score against Γ . Synthesis across candidates is NOT the default path.
3. For each consecutive pair, embed the tail window of the left and the head window of the right and compute cosine similarity.
 - $\text{sim} \geq \tau_{\text{sem}} \rightarrow$ splice directly.
 - $\text{sim} < \tau_{\text{sem}} \rightarrow$ generate a transition bridge with the local model, given both boundaries and Γ .
4. Record every seam: the pair, the similarity, and the path taken.

τ_{sem} MUST be calibrated, never fixed. Calibration uses labelled seam/non-seam pairs, maximizing F_{β} with $\beta < 1$, and **MUST** be re-derived whenever the embedding model changes. An implementation shipping a hard-coded cosine threshold is non-conformant. Rationale: contextual embedding spaces are anisotropic, cosine values are not comparable across models, and no canonical threshold exists in the literature.

14b. Consensus by multiple alignment

Section 14 resolves *different* fragments at *different* positions. This section resolves k replicas of the *same* micro-task.

1. For a micro-task of criticality $\kappa > 1$, the orchestrator **MUST** resolve the replicas by multiple alignment at semantic-unit granularity. It **MUST NOT** simply concatenate the replicas, and **MUST NOT** pick one arbitrarily.
2. Replicas **MUST** be dispatched to nodes of different `model_family` wherever the candidate pool allows it. The response **MUST** report which families contributed.
3. The response **MUST** include a per-unit agreement score, and **MUST** label each unit HIGH, MEDIUM or LOW against the calibrated thresholds α_{high} and α_{low} .
4. **α_{high} and α_{low} MUST be calibrated per embedding model, exactly as τ_{sem} is.** An implementation shipping hard-coded agreement thresholds is non-conformant.
5. Units scoring below α_{low} **MUST** be surfaced to the user as low-confidence regions. Suppressing them is non-conformant.
6. **An implementation MUST NOT present an agreement score as an accuracy or truthfulness score.** Agreement among replicas is evidence of convergence, not of correctness; models sharing training data share errors.
7. **Splitting an atomic request into partial sub-requests is NOT a supported operation.** An orchestrator that does so is non-conformant. Rationale: it removes information prior to sampling, and redundancy cannot recover information that was removed before sampling.

15. Sensitivity lanes

Lane	Criterion	Permitted destinations
PUBLIC	No PII, no commercial secret	Any conformant worker
SANITISABLE	PII detectable and pseudonymizable	Any conformant worker, after local pseudonymization; rehydrated locally at assembly
SENSITIVE	Health, legal, financial, identifiable	Local execution, or a worker presenting valid TEE attestation

Classification **MUST** run before any packet leaves the device. Pseudonymization mappings **MUST NOT** leave the device.

Implementations MUST NOT describe fragmentation as encryption in user-facing text. The SANITISABLE lane reduces risk; it does not eliminate it, and interfaces **SHOULD** say so.

15b. Anchor nodes

During bootstrap the network MAY include foundation-operated rented capacity.

1. Such nodes **MUST** be labelled `anchor: true` in the registry.
2. The network **MUST** publish the share of traffic served by anchor nodes.
3. **Presenting anchor-served traffic as community-served is non-conformant.**

15c. Privacy tiers and trusted swarms

Classification. Every request **MUST** be assigned a tier before planning, by a classifier that runs entirely on the requesting device. A classifier that consults the network in order to decide whether a request is private is non-conformant.

1. A **manual flag** (`--privacy=trusted`, `--privacy=local`) is authoritative and **MUST NOT** be overridden or downgraded by automatic triage.
2. **Automatic triage** MAY raise a tier on local named-entity detection of regulated classes. It **MUST NOT** lower one.
3. Implementations **MUST NOT** present automatic triage as certifying the absence of sensitive content. It raises a tier; it does not clear one.

Tiers.

Tier	Population	Transport	Permitted lanes
GLOBAL	Any conformant worker in the open registry	Authenticated worker identity; server-side TLS	PUBLIC, SANITISABLE
TRUSTED	Only nodes on the named swarm's public-key whitelist	Mutual TLS REQUIRED on every link	PUBLIC, SANITISABLE, and SENSITIVE where swarm policy and applicable law permit
LOCAL	The requesting device only	No network egress	All, unconditionally

Trusted-swarm requirements.

1. Swarm membership **MUST** be a whitelist of node public keys held by a swarm registry under a declared operator. A node's self-declaration of membership **MUST** be ignored.
2. Every link inside a trusted swarm **MUST** use mutual TLS. A link that authenticates only one endpoint is non-conformant for tier of TRUSTED.
3. The commitment of Section 13, Layer 1 remains **REQUIRED** inside a trusted swarm. Mutual TLS authenticates an identity, not the model behind it, and membership **MUST NOT** be accepted as evidence that a declared profile was served.
4. A trusted swarm MAY relax Layer 2 sampled audit and MAY reduce criticality k . It **MUST NOT** relax Layer 1.
5. A trusted swarm MAY set $k = 1$. When it does, the response metadata **MUST** report `consensus: null` with `consensus_waived_reason: "trusted_swarm_k1"`, and the client **MUST** surface the absence of a confidence map rather than render an empty or default-high one. Reporting a confidence map that was not computed is non-conformant.
6. Notwithstanding rule 5, k **MUST** be at least 2 whenever the swarm's measured loss rate p exceeds the configured tolerance ϵ , since $c_{\text{eff}} = c(1 - p)$ leaves no margin at $c = 1$.
7. The latency headroom of a trusted swarm **MUST NOT** be spent on a finer-grained partitioning scheme. A conformant implementation runs the same protocol at every tier and spends the headroom on the context budget S of Section 9.

Trust boundary. A trusted swarm relocates trust to the whitelist operator. Whoever administers the registry can admit a node; a compromised member inside the perimeter is more dangerous than an untrusted node outside it, because the redundancy that would have detected it may have been reduced under rule 5. Implementations **MUST NOT** describe a trusted swarm as removing the need to trust anybody.

16. Known residual channels

Declared rather than concealed:

1. **Timing correlation.** Fragments of one session are dispatched in a burst. The one-way `task_id` derivation does not hide this. Mitigation: jittered dispatch, at a cost in latency.
2. **Contract fingerprinting.** A distinctive Γ is itself a session identifier across the workers receiving it. Mitigation: per-worker paraphrase, at a cost in consistency.
3. **Content inference.** A single micro-prompt may be sensitive on its own regardless of the absence of global context.
4. **Sybil concentration.** See Section 17.

17. Trust model

- Workers are **untrusted** and assumed rational rather than merely faulty.
- The orchestrator is trusted by its own user only.
- The registry is **semi-trusted**: it MAY censor or bias discovery; orchestrators SHOULD use more than one and SHOULD retain a local view.
- Inside a TRUSTED swarm, workers are **authenticated and accountable but still not verified**: the whitelist bounds *who* may serve, not *what* they served. Section 13 Layer 1 therefore applies at every tier.
- Sybil resistance is not provided**. Layered mitigation only: accumulated reputation, cost of entry, sampled audit with economic penalty, and foundation-operated anchor nodes for cold start. Implementations MUST NOT claim Byzantine or Sybil resistance.

18. Credits

Non-transferable, non-premimed, expiring. Earned by a verified accepted fragment; spent by submitting a request. No secondary market and no account-to-account transfer. Balances expire on a published schedule.

```
{ "node_id": "...", "balance": 0, "earned_total": 0, "spent_total": 0,
  "expires": [ { "amount": 0, "at": "ISO-8601" } ] }
```

Fiat conversion is unidirectional: enterprises purchase capacity through a commercial service layer; volunteers do not sell credits. The design intent is to remain outside securities and crypto-asset regimes; it is a design goal, **not legal advice**, and requires counsel in the operating jurisdiction.

19. Response metadata

Every response MUST include:

```
{
  "rho_achieved": 1.47,
  "n_tasks": 6, "n_levels": 2,
  "tau_sem": 0.71, "tau_source": "calibrated:2026-08-13:e5-base",
  "seams": [ { "left": "...", "right": "...", "similarity": 0.68, "path": "bridge" } ],
  "coherence": { "entity_grid": 0.31, "seam_free_sentence_fraction": 0.94,
    "errors": { "entity_omission": 0, "duplicated_content": 1 } },
  "verification": { "commitments_checked": 6, "failures": 0, "audited": 1 },
  "consensus": {
    "k": 3, "families": [ "...", "...", "..."], "mean_agreement": 0.0,
    "units": [ { "label": "HIGH|MEDIUM|LOW", "agreement": 0.0 } ],
    "low_confidence_regions": [ { "unit_index": 0, "agreement": 0.0 } ]
  },
  "coverage": { "c": 0.0, "p_observed": 0.0, "c_eff": 0.0,
    "expected_uncovered_fraction": 0.0 },
  "routing": { "tier": "GLOBAL|TRUSTED|LOCAL", "swarm_id": null,
    "classifier": "manual|auto", "mtls": false },
  "consensus_waived_reason": null,
  "nodes": [ { "family": "...", "version": "...", "role": "worker" } ]
}
```

Omitting coherence is non-conformant. Omitting consensus when $k > 1$ is non-conformant. Omitting routing is non-conformant. When consensus is null because the swarm reduced k to 1, consensus_waived_reason MUST state why (Section 15c, rule 5).

20. Error codes

Code	Meaning	Orchestrator action
E_KIND_UNSUPPORTED	Worker does not serve this kind	Re-dispatch elsewhere; not a fault
E_BUDGET_EXCEEDED	max_tokens above node policy	Re-dispatch or re-plan
E_DEADLINE	Deadline passed	Hedge already fired; count as slow
E_COMMITMENT_UNSUPPORTED	Scheme unknown to worker	Downgrade to redundancy, or exclude
E_VERIFY_FAILED	Commitment mismatch	Exclude node; report to audit sampler
E_SCHEMA	Output violates expects	Retry once, then exclude
E_LANE_VIOLATION	Sensitive packet offered to open node	Abort the request. Implementation defect
E_TIER_VIOLATION	TRUSTED packet offered to a node absent from the swarm	Abort the request. Implementation defect

	whitelist	
E_MTLS_REQUIRED	Trusted-swarm link presented without mutual authentication	Refuse the node; report to the swarm registry
E_SWARM_UNKNOWN	swarm_id not resolvable in any configured registry	Abort; do not fall back to GLOBAL

21. Open questions for v1.0

1. The commitment scheme is specified by interface, not by construction; lsh-activation-v1 needs a normative definition or a normative reference.
2. Registry federation and censorship resistance are unspecified.
3. Credit expiry schedule and audit penalty magnitudes are unset pending the economics of a live network.
4. Contract compression is the highest-leverage open problem: because Γ is simultaneously the coherence mechanism, the privacy exposure and the dominant cost term, any reduction in $|\Gamma|$ at equal effect improves three properties at once.
5. Whether the orchestrator can be an 8B-class model at acceptable quality is unresolved and is the subject of hypothesis H3 in the whitepaper.
6. The granularity of a semantic unit — sentence versus clause — is unresolved, and it materially affects both alignment quality in Section 14b and the parameters of the coverage model.
7. The correlation between the agreement score of Section 14b and factual accuracy is unmeasured, and MUST NOT be assumed until it has been measured.
8. Trusted-swarm registry federation, key rotation and revocation latency are unspecified. Until they are, the failure mode of rule 1 in Section 15c is a stale whitelist rather than an unauthenticated one.
9. Whether a client-side named-entity triage model small enough to run on the requesting device reaches acceptable recall on regulated entity classes is unmeasured, and the manual flag exists because it is unmeasured.

Specification v0.2 (revision 2), 14 August 2026. Normative for the reference implementation in swarmby_v0/. Rationale, evidence and citations: WHITEPAPER_EN.md.